

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. The graph of g consists of two straight lines and a semi-circle. Use it to evaluate each integral.

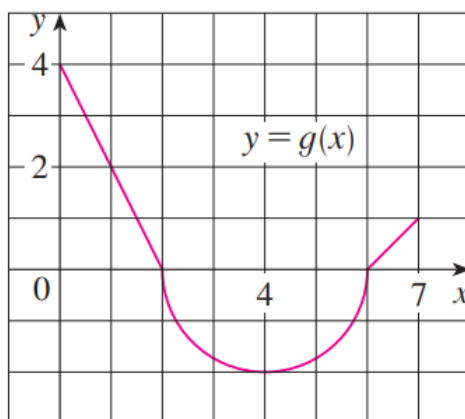
a) $\int_0^2 g(x) dx$

b) $\int_2^6 g(x) dx$

c) $\int_0^7 g(x) dx$

d) $\int_5^5 g(x) dx$

e) $\int_4^1 g(x) dx$



2. Suppose that $f(x)$ & $g(x)$ are continuous functions at all real numbers and that

$$\int_0^1 f(x) dx = 2 \quad \int_0^2 f(x) dx = -1 \quad \int_0^1 g(x) dx = 1 \quad \int_0^2 g(x) dx = 0$$

Use properties of integrals to compute each of the following.

a) $\int_0^1 [3f(x) + 2g(x)] dx$

b) $\int_1^2 g(x) dx$

c) $\int_1^0 f(x) dx$

d) $\int_1^2 [3 + 2f(x) - 5g(x)] dx$

3. Which of the following is true? How can you fully justify your answer without the use of a calculator?

$$\text{A. } \int_0^1 \sqrt{1+x^2} dx \geq \int_0^1 \sqrt{1+x} dx \quad \text{OR} \quad \text{B. } \int_0^1 \sqrt{1+x^2} dx \leq \int_0^1 \sqrt{1+x} dx$$

4. Computing the integral $\int_0^4 \frac{\pi}{16\pi + 8 \arctan(x^2 e^x)} dx$ is very difficult to do. Using properties of integrals to determine lower and upper bounds on the value of this integral.

5. Suppose that $f(x)$ is a continuous function and the average value of $f(x)$ on $[a, 2a]$ is -10 (where a is some non-zero real number).

a) Show that the area between $f(x)$ and the x -axis on this same interval is $-10a$.

b) If we know that $\int_a^0 f(x) dx = -5a - a^2$ determine a non-zero value of a such that $\int_0^{2a} f(x) dx = 0$.

c) Show that the function $f(x)$ must have a root on the interval $[0, 10]$.